

Synchronous Machine Drive Observability Analysis and Sensorless Control Design

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Abstract—Permanent magnet synchronous (PMSM) drives become very popular in many industrial applications. Their construction provides better performance and reliability than DC drives. But there is also one significant disadvantage - actual rotor position knowledge is necessary to control the drive even in the case when only speed control is required. A position sensor has to be used increasing the drive cost. Algorithms for rotor position and speed estimation from electrical quantities have been developed by many authors. This paper presents PMSM observability analysis to be able to define conditions under which it is possible to compute rotor speed and position. Following the observability analysis results a simple PMSM rotor position estimation algorithm is proposed using a MRAS concept.

Keywords—PMSM drive, observability analysis, sensorless control, MRAS

I. INTRODUCTION

Permanent magnet synchronous (PMSM) drives become very popular in many industrial applications. The main advantage of this drives is their more reliable mechanical construction comparing to DC drives. These drives can run for relatively long time without any service. On the other hand rotor position information is necessary even to be able to control drive speed. That is why in common applications a position sensor has to be used increasing significantly the drive cost.

Many algorithms for rotor position and speed estimation from electrical quantities have been developed[1], [2]. These algorithms are often based on state estimation idea (e.g. extended Kalman filter)[3], [4], [5] or position estimate adaptation using Model Reference Adaptive System structure [6], [7], [8]. Many of developed algorithms provide bad performance in the low speed region. But what is the main reason? In the case when the position and speed estimation is based on the machine model we have to deal with a question of the model state observability from system input and output values measurement. It is necessary to define conditions under which measured stator electrical quantities carry information about the mechanical rotor speed and position.

Observability analysis is a very simple task for linear dynamical systems. Unfortunately PMSM drive model is highly nonlinear and the only possibility is to apply approach introduced by Hermann and Krener [9], [10].

II. PMSM MODEL

The mathematical model of a PMSM drive is usually defined in the rotating reference frame $d - q$ by 2-phase

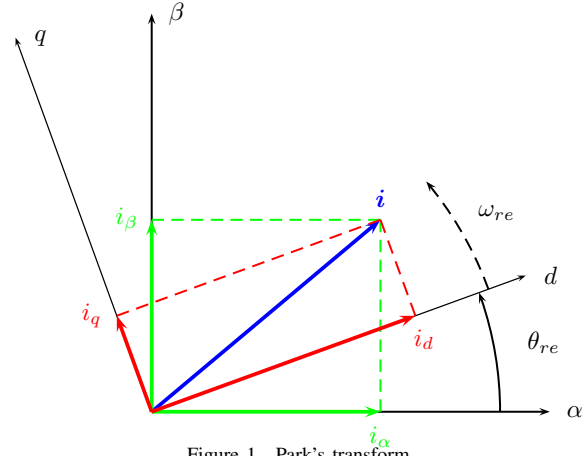


Figure 1. Park's transform

voltage equations [11]

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} L_d \frac{di_d}{dt} \\ L_q \frac{di_q}{dt} \end{bmatrix} + \begin{bmatrix} R & -\omega_{re} L_q \\ \omega_{re} L_d & R \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} K_E \omega_{re} \end{bmatrix}, \quad (1)$$

where

u_d, u_q	stator voltage components in rotating frame;
i_d, i_q	stator current components in stationary frame;
R	stator resistance
L_d, L_q	stator inductance components
K_E	EMF constant
ω_{re}	rotor angular speed (electrical speed)

The machine model can be transformed using inverse Park's transform fig.1

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} u_d \cos \theta_{re} - u_q \sin \theta_{re} \\ u_d \sin \theta_{re} + u_q \cos \theta_{re} \end{bmatrix}, \quad (2)$$

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} i_d \cos \theta_{re} - i_q \sin \theta_{re} \\ i_d \sin \theta_{re} + i_q \cos \theta_{re} \end{bmatrix}, \quad (3)$$

to stationary reference frame $\alpha - \beta$

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} L_\alpha & L_{\alpha\beta} \\ L_{\alpha\beta} & L_\beta \end{bmatrix} \begin{bmatrix} \frac{di_\alpha}{dt} \\ \frac{di_\beta}{dt} \end{bmatrix} + R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega_{re} K_E \begin{bmatrix} -\sin \theta_{re} \\ \cos \theta_{re} \end{bmatrix}, \quad (4)$$

where

$$\begin{aligned} L_\alpha &= L_0 - L \cos \theta_{re} \\ L_\beta &= L_0 - L \cos \theta_{re} \\ L_{\alpha\beta} &= L \sin \theta_{re}, \end{aligned} \quad (5)$$

$$\begin{aligned} L_0 &= \frac{L_d + L_q}{2} \\ L &= \frac{L_d - L_q}{2} \end{aligned} \quad (6)$$

and

u_α, u_β stator voltage components in stationary frame;
 i_α, i_β stator current components in stationary frame;
 θ_{re} rotor angular position (electrical angle)

Models (1),(4) describe an interior permanent magnet synchronous machine (IPMSM) with salient rotor type, where in general

$$L_d \neq L_q \quad (7)$$

If we assume a surface permanent magnet synchronous machine (SPMSM) with non-salient rotor type, then

$$L_d = L_q \quad (8)$$

and in consequent

$$\begin{aligned} L_\alpha &= L_\beta = L_d = L_q \\ L_{\alpha\beta} &= 0 \end{aligned} \quad (9)$$

System model should be in the form of state equations to be able to perform observability analysis. State equations can be easily obtained from (4) by multiplication of both sides of the equation by the inverse of the inductance matrix

$$\begin{aligned} L^{-1} &= \begin{bmatrix} L_\alpha & L_{\alpha\beta} \\ L_{\alpha\beta} & L_\beta \end{bmatrix}^{-1} \\ &= \frac{1}{L_\alpha L_\beta - L_{\alpha\beta}^2} \begin{bmatrix} L_\beta & -L_{\alpha\beta} \\ -L_{\alpha\beta} & L_\alpha \end{bmatrix} \\ \begin{bmatrix} \frac{di_\alpha}{dt} \\ \frac{di_\beta}{dt} \end{bmatrix} &= \frac{1}{L_\alpha L_\beta - L_{\alpha\beta}^2} \begin{bmatrix} L_\beta & -L_{\alpha\beta} \\ -L_{\alpha\beta} & L_\alpha \end{bmatrix} \\ &\quad \left(\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} - \omega_{re} K_E \begin{bmatrix} -\sin \theta_{re} \\ \cos \theta_{re} \end{bmatrix} \right) \end{aligned} \quad (10)$$

III. PMSM MACHINE OBSERVABILITY

A. Observability Theory

Observability analysis will be based on the non-linear systems *local weak observability* concept [9]. The advantage of this concept is that it is sufficient to prove the possibility of the system state tracking and relatively simple criterion exists to evaluate the systems local weak observability property.

Definition 1: – State X-indistinguishability
 Assume nonlinear dynamical system Σ

$$\Sigma: \begin{aligned} \frac{dx}{dt} &= f(x, u) \\ y &= h(x) \end{aligned} \quad (12)$$

where $x \in \Omega \subset \mathbb{R}^n$ is the state vector, $u \in U \subset \mathbb{R}^m$ is the vector of input values, $y \in \mathbb{R}^r$ is the output vector and f, h are nonlinear functions

$$f: \Omega \times U \mapsto \mathbb{R}^n, \quad (13)$$

$$h: \Omega \mapsto \mathbb{R}^r. \quad (14)$$

Let $x_0, x \in \Omega$ are states of the system Σ (12) and $\Theta_{x_0} u \tau, \langle t_0, t \rangle$ is a state trajectory corresponding to the control $u \tau$ at time interval $\tau \in \langle t_0, t \rangle$ and initial state $x t_0 = x_0$. States x_0, x are *indistinguishable* on the set X (X-indistinguishable), if and only if

$$\begin{aligned} \forall u \tau, \langle t_0, t \rangle \in U \times \mathbb{R} \\ \Theta_{x_0} u \tau, \langle t_0, t \rangle \subset X \times \mathbb{R} \Rightarrow \\ \Rightarrow \Sigma_{x_0} u \tau, \langle t_0, t \rangle = \Sigma_{x_1} u \tau, \langle t_0, t \rangle. \end{aligned} \quad (15)$$

Then it is possible to define the *indistinguishability relation* on the set $X \subset \Omega$ as

$$\begin{aligned} I_X: X \mapsto X \\ I_X x_0 = \{x | x \in X \wedge \forall u \tau, \langle t_0, t \rangle \in U \times \mathbb{R} \\ \Theta_{x_0} u \tau, \langle t_0, t \rangle \subset X \times \mathbb{R} \Rightarrow \\ \Rightarrow \Sigma_{x_0} u \tau, \langle t_0, t \rangle = \Sigma_{x_1} u \tau, \langle t_0, t \rangle \}. \end{aligned} \quad (16)$$

Definition 2: – State weak local observability

State of the system (12) is said to be *weakly locally observable* at the point x_0 , if and only if

$$\begin{aligned} \exists X \in \mathcal{O} \quad \forall X' \in \mathcal{O}, X' \subset X, x_0 \in X' \\ I_{X'} x_0 = \{x_0\}, \end{aligned} \quad (17)$$

where $\mathcal{O}$ is the domain of all open sets. The system (12) is said to be *weakly locally observable*, if and only if its state is weakly locally observable at every point of the state space Ω

$$\begin{aligned} \forall x_0 \in \Omega \quad \exists X \in \mathcal{O} \quad \forall X' \in \mathcal{O}, X' \subset X, x_0 \in X' \\ I_{X'} x_0 = \{x_0\}. \end{aligned} \quad (18)$$

The main advantage of this concept is that a relatively simple algebraic criterion for its evaluation exists [9], [10] while it is sufficient to define ability of the system state changes tracking.

Theorem 1: – Observability theorem

Assume nonlinear dynamical system (12) and a point from its state space $x_0 \in \Omega$. Let

$$O = \frac{\partial L}{\partial x} \Big|_{x=x_0}, \quad (19)$$

where

$$L = \begin{bmatrix} \mathcal{L}_f^0 h \\ \mathcal{L}_f^1 h \\ \vdots \\ \mathcal{L}_f^{n-1} h \end{bmatrix} \quad (20)$$

is the observability criterion matrix and $\mathcal{L}_f^k \mathbf{h}$ is the k -th order Lie derivative of the function $\mathbf{h}$ with respect to the vector field $\mathbf{f}$. If the matrix $\mathbf{O}$ has full rank

$$\text{rank}\{\mathbf{O}\} = n \quad (21)$$

then the state of the system Σ is locally weakly observable at point $\mathbf{x}_0$.

Remark 1: The Lie derivative of the function $\mathbf{h}$ with respect to the vector field $\mathbf{f}$ is given by

$$\mathcal{L}_f \mathbf{h} = \nabla \mathbf{h} \mathbf{f} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \mathbf{f} = \sum_i^n \frac{\partial \mathbf{h}}{\partial x_i} f_i, \quad (22)$$

$$\begin{aligned} \mathcal{L}_f^0 \mathbf{h} &= \mathbf{h} \\ \mathcal{L}_f^k \mathbf{h} &= \mathcal{L}_f \mathcal{L}_f^{k-1} \mathbf{h}. \end{aligned} \quad (23)$$

B. Rotor Speed and Position Observability

During the observability analysis we will make following assumptions:

- electrical parameters of the machine are known and constant
- load torque is unknown
- rotor and load inertia is unknown
- rotor speed is constant or slowly varying

According to these assumptions we have to extend the machine model (4) with two equations

$$\begin{aligned} \frac{d\theta_{re}}{dt} &= \omega_{re} \\ \frac{d\omega_{re}}{dt} &= \end{aligned} \quad (24)$$

We will apply the observability theorem to the system given by (4),(24). The equations can be rewritten into components

$$\begin{aligned} \frac{di_\alpha}{dt} &= \frac{L_0 - L \cos \theta_{re}}{L_0 - L \cos \theta_{re}} u_\alpha - \frac{L \sin \theta_{re}}{L_0 - L \cos \theta_{re}} u_\beta - \\ &\quad - \frac{L_0 - L \cos \theta_{re}}{L_0 - L \cos \theta_{re}} R i_\alpha - \frac{L \sin \theta_{re}}{L_0 - L \cos \theta_{re}} R i_\beta \\ &\quad - \frac{K_E L \sin \theta_{re} \cos \theta_{re} \omega_{re}}{L_0 - L \cos \theta_{re}} \end{aligned} \quad (25)$$

$$\begin{aligned} \frac{di_\beta}{dt} &= \frac{-L \sin \theta_{re} u_\alpha}{L_0 - L \cos \theta_{re}} - \frac{L_0 - L \cos \theta_{re}}{L_0 - L \cos \theta_{re}} u_\beta \\ &\quad - \frac{L \sin \theta_{re} R i_\alpha}{L_0 - L \cos \theta_{re}} - \frac{L_0 - L \cos \theta_{re}}{L_0 - L \cos \theta_{re}} R i_\beta \\ &\quad - \frac{K_E L \sin \theta_{re} \cos \theta_{re} \omega_{re}}{L_0 - L \cos \theta_{re}} \end{aligned} \quad (26)$$

$$\frac{d\theta_{re}}{dt} = \omega_{re} \quad (27)$$

$$\frac{d\omega_{re}}{dt} = \quad (28)$$

The equations (25)-(28) can be fitted to the structure (12) assuming

$$\mathbf{x} = \begin{bmatrix} i_\alpha \\ i_\beta \\ \theta_{re} \\ \omega_{re} \end{bmatrix}, \quad (29)$$

$$\mathbf{u} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix}, \quad (30)$$

$$\mathbf{y} = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}, \quad (31)$$

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \frac{di_\alpha}{dt} \\ \frac{di_\beta}{dt} \\ \omega_{re} \end{bmatrix}, \quad (32)$$

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (33)$$

and the state space dimension $n = 4$.

In our case the result of Lie derivative will be a vector with two components

$$\mathcal{L}_f^k \mathbf{h} = \begin{bmatrix} \mathcal{L}_f^k h_1 \\ \mathcal{L}_f^k h_2 \end{bmatrix}. \quad (34)$$

Because of the computational complexity only the first two derivatives are denoted below

$$\mathcal{L}_f^0 \mathbf{h} = \mathbf{h} = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}, \quad (35)$$

$$\mathcal{L}_f \mathbf{h} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \mathbf{f} = \begin{bmatrix} \frac{di_\alpha}{dt} \\ \frac{di_\beta}{dt} \end{bmatrix}. \quad (36)$$

As the system order is $n = 4$ it is necessary to evaluate Lie derivatives $\mathcal{L}_f^k \mathbf{h}$ up to order $k = 3$. Resulting criterion matrix has dimensions 8×8 and it is the Jacobian

$$\mathbf{O} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^1 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^1 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^1 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^1 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^2 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^2 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^3 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^3 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^3 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^3 \mathbf{h}}{\partial \omega_{re}} \end{bmatrix}. \quad (37)$$

According to the observability theorem mentioned earlier the matrix $\mathbf{O}$ have to be full-rank matrix to guarantee the system to be weakly locally observable

$$\text{rank}\{\mathbf{O}\} = 4. \quad (38)$$

Condition (38) can be tested by searching for a regular matrix constructed from any four rows of the matrix $\mathbf{O}$. There are 70 possible submatrices and we have to prove that at least one is regular. It is not an easy task, but after some experiments it is possible to find out that interesting results can be obtained using first four rows of the matrix $\mathbf{O}$. Now we will try prove that the matrix

$$\mathbf{O}_s = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \omega_{re}} \\ \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \theta_{re}} & \frac{\partial \mathcal{L}_f^0 \mathbf{h}}{\partial \omega_{re}} \end{bmatrix} \quad (39)$$

is regular. Its determinant can be computed as

$$\det\{\mathbf{O}_s\} = \frac{K_E}{L - L_0 - L_0 - L_0} \begin{vmatrix} L & Ri_\beta \cos \theta_{re} - Ri_\alpha \sin \theta_{re} - u_\beta \cos \theta_{re} \\ u_\alpha \sin \theta_{re} & K_E L_0 - L \omega_{re} \end{vmatrix}. \quad (40)$$

As the EMF constant is always K_E / ω_{re} , the matrix $\mathbf{O}_s$ regularity condition $\det\{\mathbf{O}_s\} \neq 0$ is equal to condition

$$\begin{vmatrix} L & Ri_\beta \cos \theta_{re} - Ri_\alpha \sin \theta_{re} - u_\beta \cos \theta_{re} \\ u_\alpha \sin \theta_{re} & K_E L_0 - L \omega_{re} \end{vmatrix} \neq 0. \quad (41)$$

If the analyzed drive is a SPMSM the inductances L_d and L_q have the same value and from (6) we will get $L = L_0$. In this case the observability condition (41) is reduced to

$$\omega_{re} \neq 0. \quad (42)$$

For IPMSM drive the complete condition (41) have to be evaluated. It is possible to see that the observability condition follows pattern of the Park's transform

$$\begin{bmatrix} u_q \\ i_q \end{bmatrix} = \begin{bmatrix} -u_\alpha \sin \theta_{re} & u_\beta \cos \theta_{re} \\ -i_\alpha \sin \theta_{re} & i_\beta \cos \theta_{re} \end{bmatrix}. \quad (43)$$

From (6) we can see, that

$$\begin{bmatrix} L_0 & L \\ L & L_d - L_q \end{bmatrix}. \quad (44)$$

Observability condition (41) can be transferred using (43) and (44) into form

$$L_d - L_q Ri_q - u_q = K_E L_d \omega_{re} / \omega_{re}. \quad (45)$$

The term $Ri_q - u_q$ can be isolated from IPMSM model (4) as

$$Ri_q - u_q = \omega_{re} L_d i_d - L_q \frac{di_q}{dt} - \omega_{re} K_E. \quad (46)$$

After substitution into observability condition (45) we will get

$$L_q K_E \omega_{re} - L_d - L_q L_d i_d \omega_{re} - L_d - L_q L_q \frac{di_q}{dt} \neq 0. \quad (47)$$

In most cases the rotor magnetic flux is produced only by the permanent magnet (with exception of running the drive at high speed in the field weakening region). Then the flux producing component of the stator current is controlled to be zero $i_d = 0$. The observability condition can be simplified to

$$K_E \omega_{re} \neq L_d - L_q \frac{di_q}{dt}. \quad (48)$$

It is necessary to say, that both the SPMSM observability condition (42) and IPMSM observability condition (48) are only sufficient, not required conditions to achieve drive state observability. It means that the drive state may be observable even if these conditions are not fulfilled. If these conditions are fulfilled the state observability is guaranteed.

IV. SENSORLESS CONTROL DESIGN

A. Position Estimation Algorithm

In previous section it was shown that IPMSM drives are more suitable for sensorless control applications. Unfortunately most of the produced PMSM drives are optimized to have no (or at least minimal) saliency in the stator inductance to achieve both sinusoidal stator voltage and current and to reject higher harmonic components in the stator current. That is why we will consider the case described by (8),(9) when the machine model in the stationary reference frame can be reduced to

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = L \begin{bmatrix} \frac{di_\alpha}{dt} \\ \frac{di_\beta}{dt} \end{bmatrix} + R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega_{re} K_E \begin{bmatrix} -\sin \theta_{re} \\ \cos \theta_{re} \end{bmatrix} \quad (49)$$

where L is the stator inductance. Applying integration operation to (49) we will get

$$\int \left(\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \right) dt = L \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + K_E \begin{bmatrix} \cos \theta_{re} \\ \sin \theta_{re} \end{bmatrix}. \quad (50)$$

The left side of (50) can be considered to be the magnetic flux voltage model

$$\begin{bmatrix} \Psi_{u\alpha} \\ \Psi_{u\beta} \end{bmatrix} = \int \left(\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \right) dt, \quad (51)$$

$$\begin{bmatrix} \frac{d\Psi_{u\alpha}}{dt} \\ \frac{d\Psi_{u\beta}}{dt} \end{bmatrix} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}, \quad (52)$$

while the right side of (50) is the magnetic flux current model

$$\begin{bmatrix} \Psi_{i\alpha} \\ \Psi_{i\beta} \end{bmatrix} = L \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + K_E \begin{bmatrix} \cos \theta_{re} \\ \sin \theta_{re} \end{bmatrix}. \quad (53)$$

Using the complex number notation $\mathbf{u} = u_\alpha + j u_\beta$, $\mathbf{i} = i_\alpha + j i_\beta$ and $\Psi_u = \Psi_{u\alpha} + j \Psi_{u\beta}$, $\Psi_i = \Psi_{i\alpha} + j \Psi_{i\beta}$ the

magnetic flux model can be rewritten into

$$\frac{d\Psi_u}{dt} \Delta u - Ri, \quad (54)$$

$$\Psi_i \Delta Li \Delta K_E \Delta \theta_{re}. \quad (55)$$

If we assume the machine parameters to be known, both models should provide the same value of the magnetic flux. The difference between Ψ_u and Ψ_i occurs only as a consequence of the error in the rotor position θ_{re} knowledge. If the difference between space vectors calculated by the voltage and current models are small, it can be shown [12], that the errors in the flux size $\Delta\Psi_u$ and the angle $\Delta\theta_{re}$ can be calculated as

$$\Delta\Psi_u \approx \Psi_i - \Psi_u, \quad (56)$$

$$\Delta\theta_{re} \approx \frac{\Im\{\Psi_u \bar{\Psi}_i\}}{\Psi_u \bar{\Psi}_i}. \quad (57)$$

While the equation (57) seems to be very simple, it can cause implementation problems. The first problem is that the denominator in (57) is close to zero, during the algorithm initial condition. The other problem is the use of division by a variable expression, which is difficult to implement on simple low-cost controllers. That is why we will try to find better expression for position error calculation. We will assume, that the magnetic flux space vector calculated by the voltage model is correct and it is equal to the correct value computed by the current model

$$\Psi_u \Delta Li \Delta K_E \Delta \theta_{re}, \quad (58)$$

while the value calculated by the current model is rotated by the error angle $\Delta\theta_{re}$

$$\Psi_i \Delta Li \Delta K_E \Delta \theta_{re} \Delta^{-j\Delta\theta_{re}}. \quad (59)$$

Now we can get

$$\begin{aligned} \Psi_u \bar{\Psi}_i \Delta (Li \Delta K_E \Delta \theta_{re}) (L\bar{i} \Delta K_E \Delta^{-j\Delta\theta_{re}} \Delta \Delta \theta_{re}) \Delta \\ \Delta L \Delta i \Delta K_E \Delta \theta_{re} \Delta Li \Delta K_E \Delta^{-j\Delta\theta_{re}} \Delta \Delta \theta_{re} \Delta \\ \Delta L \bar{i} K_E \Delta \theta_{re}, \end{aligned} \quad (60)$$

$$\begin{aligned} \Im\{\Psi_u \bar{\Psi}_i\} \Delta K_E \Delta \sin \Delta \theta_{re} \Delta \\ \Delta L K_E \Im\{i\} \cos \Delta \theta_{re} - \Delta \theta_{re} \Delta \\ - L K_E \Re\{i\} \sin \Delta \theta_{re} - \Delta \theta_{re} \Delta \\ - L K_E \Im\{i\} \cos \theta_{re} \Delta L K_E \Re\{i\} \sin \theta_{re}. \end{aligned} \quad (61)$$

If the error in the rotor position is small $\Delta\theta_{re} \rightarrow \Delta$ it is possible to assume

$$\theta_{re} \Delta \Delta \theta_{re} \approx \theta_{re}, \quad (62)$$

$$\sin \Delta \theta_{re} \approx \Delta \theta_{re}. \quad (63)$$

Using (62),(63), equation (61) can be simplified to

$$\Im\{\Psi_u \bar{\Psi}_i\} \approx K_E \Delta \theta_{re} \quad (64)$$

and thus

$$\Delta \theta_{re} \approx \frac{\Im\{\Psi_u \bar{\Psi}_i\}}{K_E}. \quad (65)$$

Rotor position estimate θ_{re} can be easily calculated using value computed in (65) as an input of a PI controller with

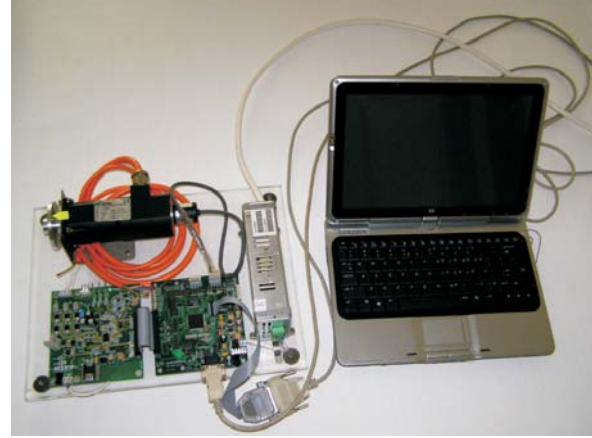


Figure 2. Experimental PMSM drive system

an additional integration [12]. Its transfer function can be described as

$$F_{\theta_{re}} \Delta \Delta \Delta \frac{\theta_{re} \Delta \Delta}{\Delta \theta_{re} \Delta \Delta} \Delta \frac{k_{\theta} s \Delta k_{\omega}}{s \Delta}, \quad (66)$$

where k_{θ}, k_{ω} are design parameters. At the same time, flux error computed from (56) can be used to correct inaccuracy of the voltage model caused by changes in stator resistance and error in the initial condition knowledge. The correction can be made by modifying the voltage model (54) into

$$\frac{d\Psi_u}{dt} \Delta u - Ri \Delta \delta_{\Psi}, \quad (67)$$

where the correction δ_{Ψ} is computed by a couple of PI controllers with the transfer function matrix

$$F_{\Psi} \Delta \begin{bmatrix} \frac{k_1 + k_2 s}{s} & \Delta \\ \Delta & \frac{k_1 + k_2 s}{s} \end{bmatrix}, \quad (68)$$

$$\delta_{\Psi} \Delta \Delta \Delta F_{\Psi} \Delta \Psi_u \Delta \Delta \quad (69)$$

where k_{Δ}, k_{Δ} are design parameters.

B. Algorithm Implementation and Testing

The proposed speed estimation algorithm was implemented using Freescale 56F8346 chip. This chip is developed for motor control applications and is equipped with all necessary peripheral units. The algorithm was tested on the real electromechanical system shown on Fig. 2. The system consist of Freescale 56F8346 controller board, Freescale low voltage power-stage and TG-Drives low voltage (12V) PMSM drive with power 100W.

The comparison of the real and estimated rotor position is shown on Fig. 3. The drive was controlled using a position sensor during this experiment, while the position was estimated in parallel computation. It can be seen, that the algorithm is able to estimate the rotor position even in the case of a big error in the initial rotor position knowledge.

Speed estimation performance is shown on Fig. 3. The algorithm can track the motor speed except in low speed region. This corresponds with results of the observability analysis. In low speed region other position and speed estimation methods should be used (eg. saturation effects detection).

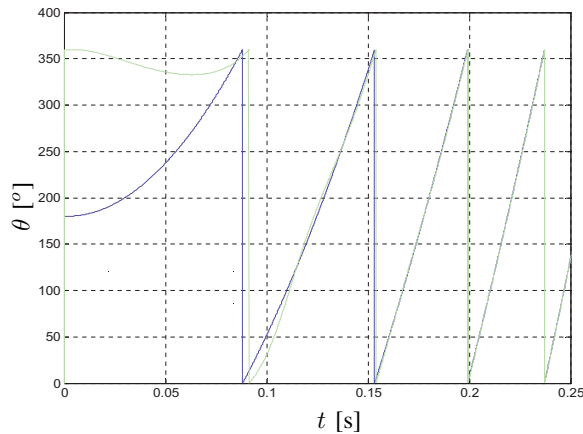


Figure 3. Real (blue) and estimated (green) position during the drive start-up

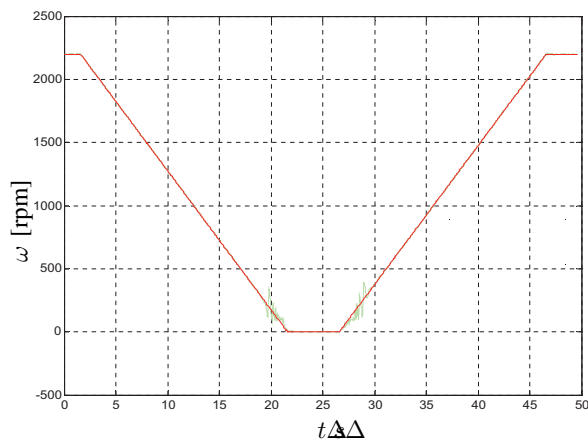


Figure 4. Real (red) and estimated (green) mechanical speed

V. CONCLUSIONS AND FUTURE WORK

A. Conclusions

The results of the PMSM state observability proved, that rotor position and speed can be computed from stator electrical quantities measurement. It has been shown that in the case of surface permanent magnet machine, the observability condition is given by (42). The possibility of the rotor speed and position estimation is guaranteed for nonzero rotor speed only. This property of SPMSM is known causing troubles in sensorless control in low speed region and estimation of the initial rotor position. On the other hand observability condition (48) valid for IPMSM drives promise possibility of the rotor position estimation even at zero rotor speed. The state observability can be achieved by applying changes to stator current allowing relatively easy initial rotor position estimation. This is very important as it is possible to design a IPMSM drive sensorless control without performing initial rotor alignment procedure.

Simple position estimation algorithm has been proposed and implemented on a real control system. The algorithm was proved in the testing on a real PMSM drive.

B. Future Work

The research group at the Brno University of Technology performs intensive research in the field of AC induction machine [13] and synchronous machine sensorless control. The current paper presents sufficient condition for PMSM states observability. The future research will be aimed especially to necessary observability conditions analysis. The purpose of the research is to clearly define requirements on the machine control to preserve its states observability. Observability analysis of other electrical drives types will be also carried out [14]. We are also going to deal with design with modern AC drives control algorithms, especially in the field of the predictive control.

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